Ohm's law: $V=IR$

Resistance is directly proportional to voltage at constant temperature

Series and Paralel circuits

- Series:

$I_1 = I_2 = I_3$ (current same)

$V_{total} = V_1 + V_2 + V_3$

$R_{total} = R_1 + R_2 + R_3$

- Parallel:

$V_1 = V_2 = V_3$

$I_{total} = I_1 + I_2 + I_3$

$$\frac{1}{R_{total}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

Basics of Electronics notes

Electronics and charge

Key formula: $Q = I \cdot t$

$Q \rightarrow$ charge (columbs, C)

$I \rightarrow$ current (Amperes, A)

$t \rightarrow$ time

- Conduction electrons - free electrons moving around atom cores in metals
- Conventional current - flows $+ \rightarrow -$
  (opposite to electron flow)

Potential Difference

Key formula: $V = \frac{W}{Q}$

V - voltage (Volts)

W - energy (Joules)

Q - charge (Columbs)

Kirchhoff's First Law (KCL)

Statement: Sum of currents entering a junction = sum of currents leaving

Formula ⇒ $I = I_1 + I_2 + I_3$

Key points:

- Consequence of conservation of charge
- Current does not increase or decrease at junction
- Junction = point ~~where ≥3~~ where ≥3 circuit paths meet.
- Branch → path connecting two junction
- Series circuit → current same at every point
- Parallel circuit → current divides at junctions, each branch carries different value

Kirchhoff's Second law (KVL)

Statement: Total EMF in a closed loop = sum of all potential differences across components.

Formula: $E_1 + E_2 = V_1 + V_2$

Key points:

- Consiquences of conservation of energy
- Energy in = energy out
- Series circuit → total p.d splits across components propotional to resistance
- Parallel circuit → p.d same across each closed loop.
- Each closed loop acts as independent series circuit
- if one branch breaks ops → current still flows through others

EMF and Internal Resistance

Key formulas:

$E = V + Ir$, $E = I(R + r)$,

Lost volts $= Ir$

Where:

E = EMF

V = terminal p.d.

I = current

R = external resistance

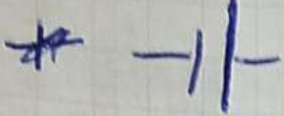
r = internal resistance

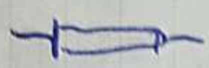
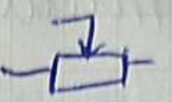
EMF vs P.D:

EMF → energy given to circuit by source

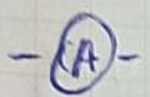
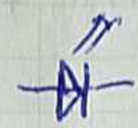
P.D → energy taken from by components

Some circuit symbols

 Battery / cell  diode

 Resistor  Potentiometer

 Switch  LDR

 Ammeter  LED

 Voltmeter

Potential Divider

Key formula: $V_{out} = V_{in} \cdot \frac{R_2}{R_1 + R_2}$

How it works:

- Two resistors in series across ~~a supply~~ a supply
- Output voltage taken across R_2
- changing R_1 or R_2 changes V_{out}

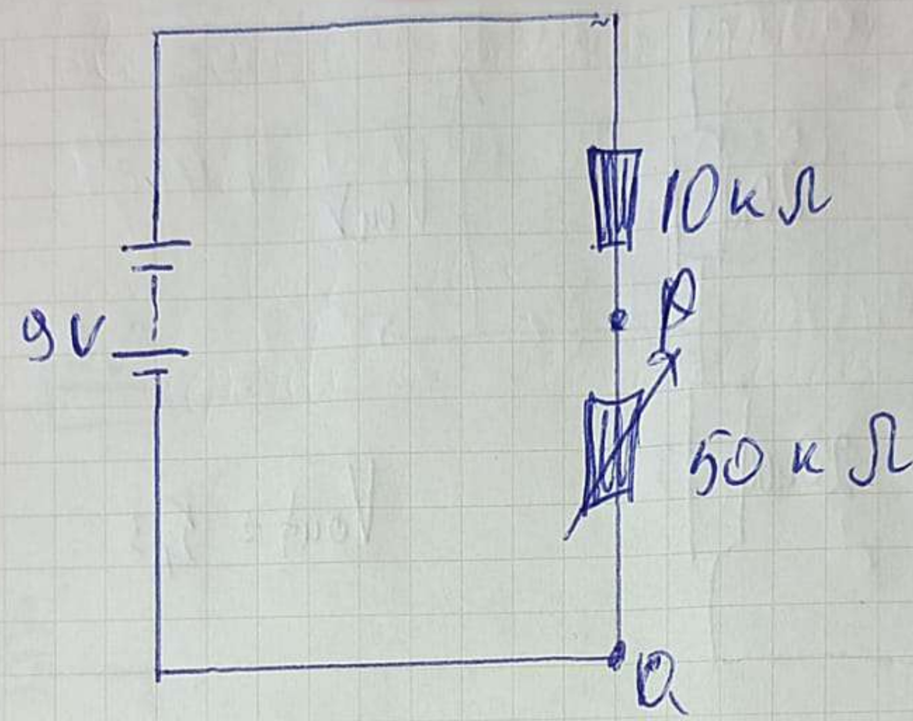
Variable resistor (potentiometer)

$R_1 = 0 \rightarrow V_{out} = 0$

$R_1 = max \rightarrow V_{out} = max$

Range depends on resistor values

3)



V_2 = potential difference between the terminals P and Q.

V_2 max:

$$V_2 = 9 \cdot \frac{50}{50+10} = 7,5\,V$$

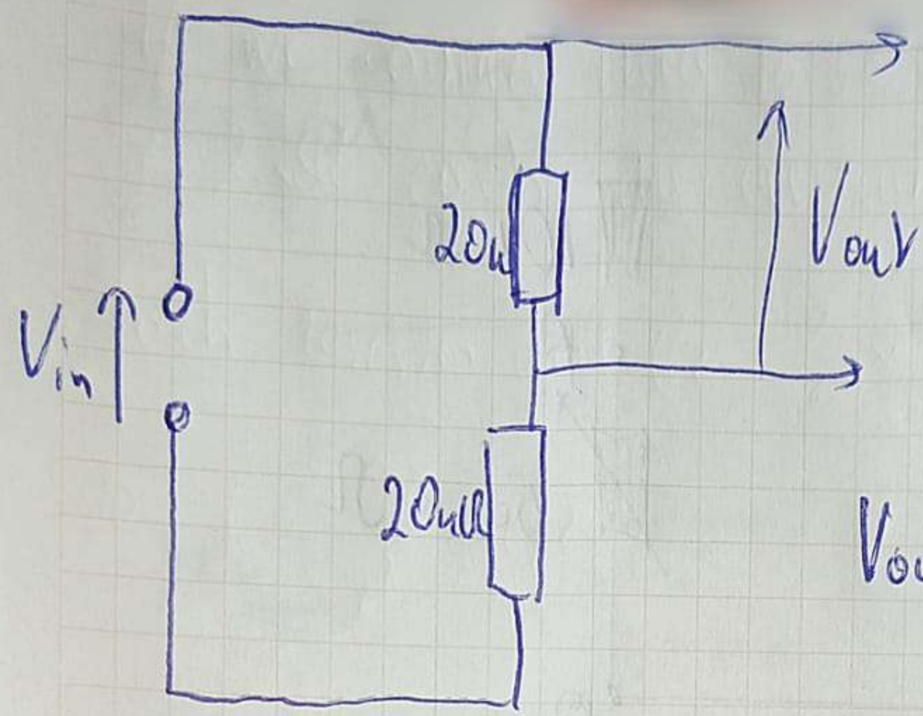
min = 0V

max = 7,5V

V_2 min:

$$V_2 = 9 \cdot \frac{0}{0+10} = 0\,V$$

[redacted]

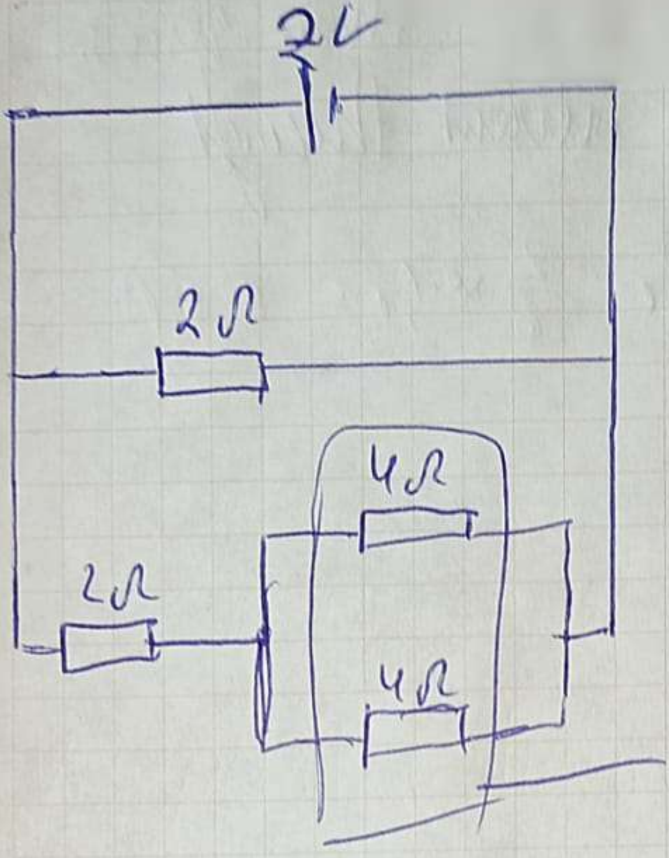


$V_{out} = 5,3$

$$V_{out} = V_{in} \cdot \frac{R_2}{R_1+R_2} \qquad \cancel{V_{in} = \frac{R_2 \cdot V_{out}}{R_1+R_2}}$$

$$V_{in} = \frac{R_1+R_2}{R_2} \cdot V_{out} = \frac{20+20}{20} \cdot 5,3 = 10,6$$

If $R_1 = 12\,k\Omega$
$R_2 = 20\,k\Omega$ $\rightarrow$ $V_{in} = \frac{12+20}{20} \cdot 5,3 = 8,5$



$$\frac{1}{R_{1+2}} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{4} + \frac{1}{4} =$$

$$\frac{1}{2}$$

$$R_{1+2} = 2\,\Omega$$

$$\cancel{R_1 +} R_{1+2+3} = 2+2 = 4\,\Omega$$

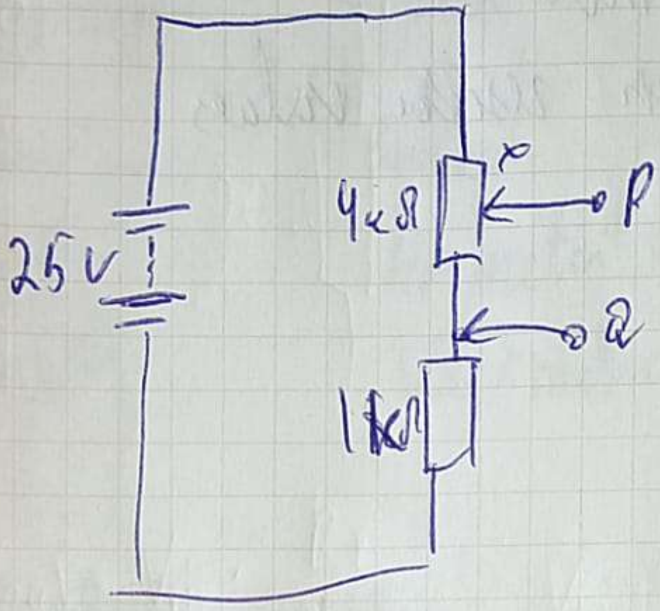
$$\frac{1}{R_{total}} = \frac{1}{R_{1+2+3}} + \frac{1}{R_4} = \frac{1}{4} + \frac{1}{2} = \frac{3}{4} =$$

$$R_{total} = \frac{4}{3}\,\Omega$$

$$I = \frac{V}{R} = \frac{2V}{\frac{4}{3}} = 1{,}5\,A$$

Practical Questions

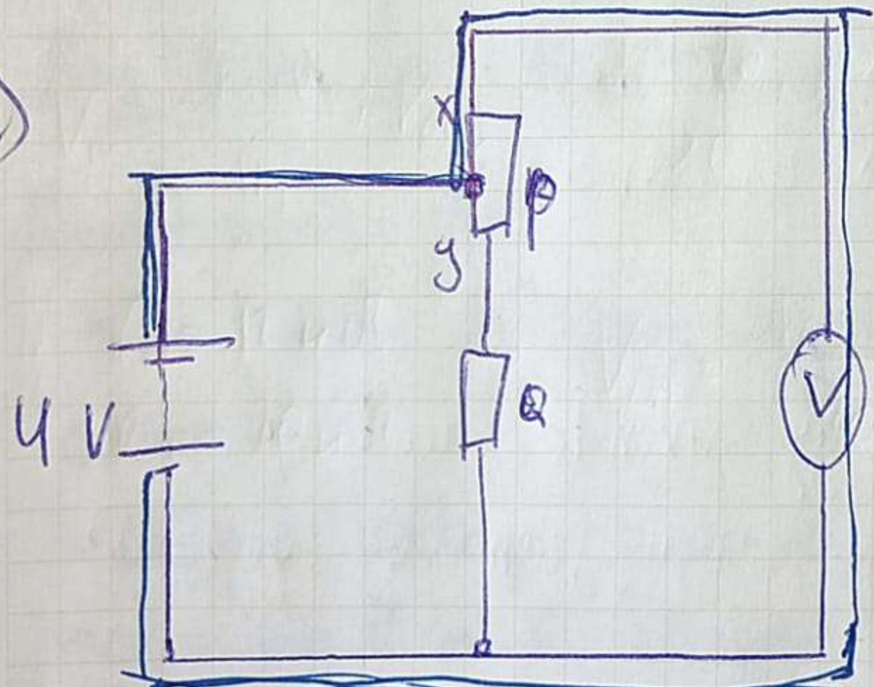
1)



$V_2 = 25 \cdot \left(\frac{4}{1+4}\right) = 20V$

$V_2 = 25 \cdot \left(\frac{0}{1+0}\right) = 0V$

2)



Voltmeter always reads 4 V.

~~Becaue~~ Becuase current never flows on Q, if we consider these to be three resistors of 2 Ω each, the current through each resistor will be $\frac{2}{6} \cdot 1{,}5 = 0{,}5$ A

so $I = \frac{0{,}5}{2} = 0{,}25$